

34 2026-05-04 | Week 16 | Lecture 34

Based on chapters 12.1 and 12.2 in the textbook

Suppose we are given data (t_i, b_i) , $i = 1, \dots, n$. The **simple linear regression problem** is to find the line $y = C + Dt$ which best fits the data.

In this problem, we understand the “line of best fit” to be the line $y = C + Dt$ in which C and D are chosen such that the **sum of squared errors**

$$e_1^2 + \dots + e_n^2$$

is minimized. Here, the e_i ’s are the entries of the **residual vector** e , defined as:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = b - Ax$$

where

$$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_n \end{bmatrix}$$

In the last lecture, we saw that minimizing the sum of squared errors is equivalent to minimizing the length of the vector $\mathbf{e}$.

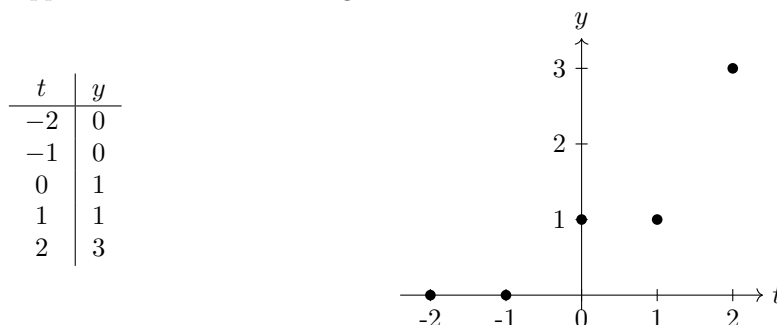
We also saw two approaches to finding the best values for C and D . The first approach involved calculus. The second approach used linear algebra, which is what we’ll focus on today. Recall the following definition:

Definition 146 (Normal equations). The **normal equations** are

$$A^\top Ax = A^\top b.$$

The linear algebra approach is easy to summarize: to find the best fit line $y = C + Dt$, set up the equation $A^\top Ax = A^\top b$ and solve it for $x = \begin{bmatrix} C \\ D \end{bmatrix}$.

Example 147. Suppose we have the following data:



Question: Find the best-fitting line $y = C + Dt$.

Solution: Consider the equation

$$Ax = b$$

where

$$A = \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

Since the data points don't lie on a line, there is no solution (i.e., no values of C and D which make this equation true). So instead, we set up the normal equations

$$A^T Ax = A^T b$$

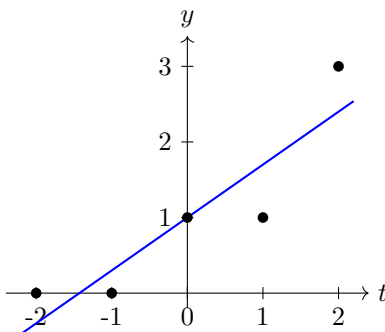
which are

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 3 \end{bmatrix}.$$

This simplifies to

$$\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

From this we conclude that $C = 1$ and $D = 7/10$. Hence, our best-fitting line is $y = 1 + .7t$.



End of Example 147. $\square$

Example 148. Question: Find the best fitting line to the data

t	y
0	1
1	2
2	2
3	4
4	5

Solution: We have

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 4 \\ 5 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} C \\ D \end{bmatrix}.$$

Therefore the normal equations $A^T Ax = A^T b$ are

$$\begin{bmatrix} 5 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 14 \\ 38 \end{bmatrix}$$

or equivalently

$$\begin{cases} 5C + 10D = 14 \\ 10C + 30D = 38 \end{cases}$$

Solving this system gives $C = .8$ and $D = 1$. Therefore the best-fit line $y = C + Dt$ is

$$y = .8 + t.$$

End of Example 148. $\square$

Theorem 149. *The values C and D which minimize $e_1^2 + \dots + e_n^2$, are given by the solution $x = \begin{bmatrix} C \\ D \end{bmatrix}$ to the normal equations*

$$A^\top Ax = A^\top b,$$

where

$$A = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_n \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Proof. I tried to explain this geometrically. If x is the solution to the normal equations, then $e = b - Ax$ is orthogonal to $\text{CS}(A)$, which guarantees that the length of the residual vector is minimized. $\square$